Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 30-06-2026
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Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Technical Architecture

User (Web Browser)
|
HTML/CSS/JavaScript Interface
|
Flask Web Application (Python)
|
Tableau Embedded Dashboard
|
Electricity Consumption Dataset (CSV)
|
Data Processing & Analysis
|
Interactive Charts, Maps & Filters

Infrastructure: Local System
Data Storage: Local CSV Dataset
External Interface: Tableau Public Embed
Machine Learning: Not Applicable (Current Version)

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for dashboard	HTML, CSS, JavaScript
2	Application Logic-1	Backend processing	Python, Flask
3	Application Logic-2	Dashboard visualization	Tableau Public
4	Application Logic-3	Data processing	Pandas
5	Database	Dataset storage	CSV File (Local)
6	Cloud Database	Not used	N/A
7	File Storage	Electricity dataset	Local File System
8	External API-1	Tableau Embed	Tableau Public
9	External API-2	Not Applicable	N/A
10	Machine Learning	Future	N/A

		enhancement	
11	Infrastructure	Application deployment	Local Server

Local Server Configuration

OS: Windows

Language: Python 3.x

Framework: Flask

Visualization: Tableau Public

Browser: Chrome/Edge

Dataset: CSV

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Backend web framework	Python, Flask, Pandas
2	Security Implementations	Basic application security	Input Validation, HTTPS (deployment)
3	Scalable Architecture	Modular architecture	Flask + Tableau
4	Availability	Accessible through local/web deployment	Local Server/Tableau Public
5	Performance	Fast dashboard rendering	Tableau Optimization, Pandas